MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 1B54

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pre-TR, integral cavity, broadband, frequency range 3,400 to 3,700 MHz.

ABSOLUTE RATINGS:

Parameter:	Incident power	Storage temp	Alt
Unit:	kw	°C	ft
Maximum:	750	+100	10,000
Minimum:	100	-55	---

PHYSICAL CHARACTERISTICS:

See figure 1
Weight: 1 pound (approx)
Mounting position: Any
Mount: Per Drawing 153-JAN

TEST CONDITIONS:

Parameter:	po	prr	tp	σ'
Unit:	kw	pps	μs	---
Tolerance:	±10%	---	±0.1	---
Test condition 1:	50(max)	1,000	1.0	1.1
Test condition 2:	200	1,000	1.0	1.1
Test condition 3:	750	400	1.0	---

Frequency		
F	MHz	±
F1	3,400	0.5%
F2	3,550	5.0%
F3	3,700	0.5%

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 202640-15T12:47Z
Check the source to verify that this is the currentversion before use.

METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4452	Spike-leakage energy	-	2	F = F2	Ws	0.8×10^{-4}	15×10^{-4}	Joules
4488	Arc loss	-	1	F = F2	La	---	0.8	dB
4486	Firing time	-	1	F = F2	t	---	10	sec
1027	Temperature cycling (nonoperating)	3	-		---	---	---	---
	Quality conformance inspection, part 2							
4473	Low-level VSWR	4	-	σ' = 1.05 (max);				
				F = F1	σ	---	1.20	---
				F = F2 ± 0.5%	σ	---	1.10	---
				F = F3	σ	---	1.20	---
4416	Insertion loss	4	-	F = F2 ± 0.5%; σ' = 1.05(max)	Li	---	0.20	dB
4471	Recovery time	2	3	F = F2	t	---	15	µs
4401	Dielectric material strain	-	-		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	-	3	Group C: F = F2	t	500		hrs
---	Life-test end points:							
4471	Recovery time	2	3	F = F2	t	---	30	µs
4488	Arc loss	4	1	F = F2	La	---	0.9	dB
4021	Degradation due to vibration	5	-		---	---	---	---
4474	High-level VSWR	5	1	F = F2	σ	---	1.2	---
1027	Temperature cycling life test	5	-	10 cycles (min)	---	---	---	---
---	Temperature cycling life-test end points:							
4488	Arc loss	5	1	F = F2	La	---	0.9	dB
4471	Recovery time	5	3	F = F2	t	---	20	µs

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. The loss of signal in the tube at the specified time after start of the transmitting pulse shall not be greater than 3.0 dB in the excess loss at 800 to 1,000 μs after the pulse. There shall be no arcing between the tube and the contact gasket at the specified power level. To supply additional protection to the crystal, use either a fast recovery time 1B27 as shown on figure 2, or isolate the mixer from the pre-TR by 10 dB of lossy cable.

3. Test each tube for compliance with arc loss requirements following this test.

4. Low-level measurements to be performed using mount per Drawing 220-JAN.

5. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - EC

(Project 5960-3026)

Source: https://assist.dla.mil -- Downloaded: 202610-15T12:47Z
Check the source to verify that this is the current version before use.

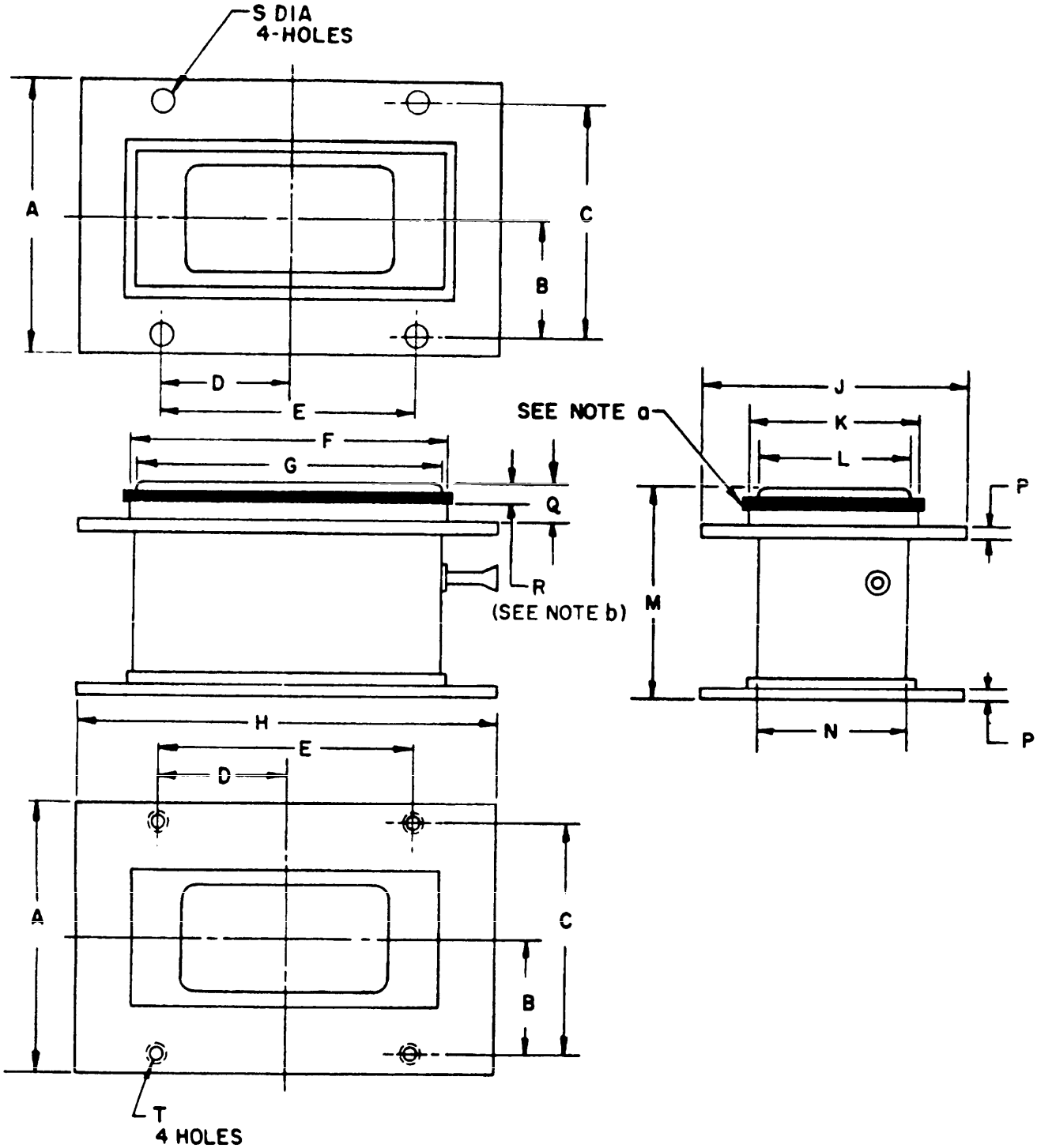


FIGURE 1. Outline drawing of electron tube type 1B54.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, Part 1 (see note 1)	
G	2.992 (76.00)	3.008 (76.40)
M	2.040 (51.82)	2.080 (52.83)
Q	.347 (8.81)	.377 (9.58)
	Quality conformance inspection, Part 2	
C	2.240 (56.90)	2.260 (57.40)
E	2.490 (63.25)	2.510 (63.75)
F	3.203 (81.36)	3.233 (82.12)
K	1.703 (43.26)	1.733 (44.02)
L	1.492 (37.90)	1.508 (38.30)
R	.165 (4.19)	.195 (4.95)
S	.214 (.5.44)	.224 (5.69)
	No. 10-32 UNC-2B	

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses
	Reference dimensions
A	2.625 (66.68)
B	1.125 (28.58)
D	1.250 (31.75)
H	4.125 (104.78)
J	2.625 (66.68)
N	1.500 (38.10)
P	.125 (3.18)

NOTES:
a. Gasket per drawing 189 JAN. Gasket to be firmly attached for a minimum of 75 per cent of contact area.
b. Dimension R shall be measured prior to the attachment of the gasket to the tube.
c. Nickel plate 100MSI, or equal.

FIGURE 1. Outline drawing of electron tube type 1B54 - continued.

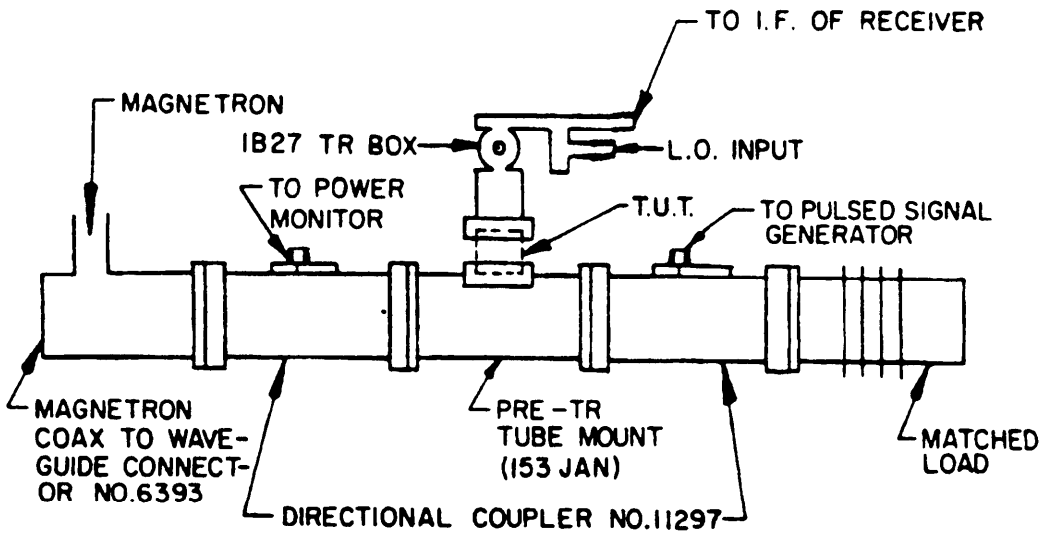


FIGURE 2. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:47Z
Check the source to verify that this is the current version before use.